

My Understanding of Machine Learning Types

- **Supervised Learning:** You train a model with labeled examples. For each example, you give the input features (x) and the correct output (y). The model then studies the link between x and y. After that, it uses what it learned to produce predictions for new, unseen inputs.
- **Unsupervised Learning:** Looking for patterns, structures, or clusters in the input data (x) by itself. This does it without any fixed labels or target outputs (y).
- **Reinforcement Learning:** You can train an agent by letting it try things and learn from the results. The agent takes actions in an environment, then it gets feedback. The goal is to end up with higher total rewards over time and fewer penalties along the way.

My Analysis of the Wine Classification Project

- **Best performing model:** The best performing label was **Logistic Regression**. It's accuracy is 88.9%
- **Why do you think this model performed better?:** Logistic Regression performed better because the relationship between the four selected features (alcohol, malic_acid, ash, and alcalinity_of_ash) and the wine classes is relatively linear. Decision Trees with a strict depth constraint (max_depth=3) can create coarse, axis-aligned decision boundaries that struggle with fine numerical boundaries. Furthermore, Logistic Regression achieved perfect precision and recall (1.00) for class_0 and only misclassified 4 samples in total (1 in class_1 and 3 in class_2).
- **What would you try next to improve performance?**
 1. **Feature Scaling:** Apply StandardScaler to normalize features, especially when including high-magnitude variables like proline.
 2. **Feature Selection & Expansion:** Use all 13 features instead of limiting the inputs to 3 or 4, or apply feature selection techniques like SelectKBest.
 3. **Hyperparameter Tuning:** Experiment with different max_depth settings or criterion types for the Decision Tree, or tune the regularization strength parameter (C) in Logistic Regression.

Real-World Application Ideas

- **Industry of Interest:** Healthcare / Medical Diagnostics
- **ML Problem:** Automated detection and classification of cardiac arrhythmias from patient ECG monitoring.
- **Type of ML:** Supervised Learning

- **Data Needed:** Time-series electrocardiogram (ECG) heart rate signal data paired with verified diagnostic annotations provided by cardiologists.

Key Learnings

- **Most important concept learned:** I learned to split the data into a training part (80%) and a test part (20%). I used stratification in `train_test_split` so the test set stays representative. This helps me judge how the model will do on new, unseen data and it reduces overfitting.
- **Most challenging part:** Picking the right features was challenging for me. For instance, I moved from the original 4 features, which gave 88.9 percent accuracy, to a new set of three: ['alcohol', 'color_intensity', 'proline']. After that change, Logistic Regression dropped to 83.3 percent accuracy.
- **Questions for further exploration:** What is the best way to deal with class imbalance when the classes are not evenly spread? Also, what steps can stop model quality from falling off once the model is used on real data that comes in as a steady stream?